

Article

Soybean Yield Modeling and Analysis with Weather Dynamics in the Greater Mississippi River Basin

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Abstract: Accurate crop yield prediction and modeling are essential for ensuring food security, optimizing resource allocation, and guiding policy decisions in agriculture, ultimately benefiting society at large. With the increasing threat of weather change, it is important to understand the impacts of weather dynamics on agricultural productivity, particularly for crucial crops like soybeans. This study considers the study area of the Greater Mississippi River Basin, where most soybeans are typically planted, with a large variety of weather across from the North to the South in the US. Leveraging the greenness and density measured by the normalized difference vegetation index (NDVI) from NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) satellite images, along with weather variables including mean precipitation, minimum temperature, and maximum temperature, we aim to uncover the relationships between these variables and soybean yield for different geographical and weather regions. Our analysis focuses on the four weather regions within the US: Very Cold, Cold, Mixed Humid, and Hot Humid, where most soybeans are planted in the Mississippi River Basin. The findings reveal that soybean yield in the Cold and Very Cold regions is positively correlated with minimum temperatures, whereas in the Mixed Humid and Hot Humid regions, negative correlations between maximum temperatures and yields are found. We identify a significant positive correlation between precipitation and soybean yield across all regions. In addition, the NDVI shows significant positive correlations with the soybean yield. Both linear and nonlinear regression models, including support vector machine and random forest models, are trained with remotely sensed data and weather data, showing a reliable and improved crop yield prediction. The findings of this study contribute to a better understanding of how soybean yield responds to climatic variations and will help the national agricultural management system in better monitoring and predicting crop yield when facing the increasing challenge of weather dynamics.

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Keywords: soybean yield prediction; yield modeling; weather dynamics; remote sensing

1. Introduction

Agriculture is a critical sector that sustains human life and develops nations by providing food and raw materials [1]. Even though food production has significantly increased in the last half century, there are still many people who do not have enough food to eat, making them lack sufficient energy and protein in their diets [2]. In the face of global weather dynamics, understanding how various climatic factors affect crop yields is paramount for ensuring food security and optimizing agricultural practices. Accurate and timely crop yield predictions are therefore critical for policymakers, allowing them to make educated

decisions regarding food import and export policies. This, in turn, is vital for strengthening food security and ensuring that food supply systems can adapt to the evolving challenges posed by weather dynamics.

Recent advancements in crop yield modeling and prediction have gained considerable attention due to the application of various methods, experiments, and data sources. Empirical models have demonstrated the reliability of remotely sensed images for estimating canopy cover fractions [3] and utilizing crop physiological indices, such as leaf area index [4] and crop greenness index [5], for effective yield prediction. Remote sensing technology enables accurate and timely yield predictions throughout the growing season by leveraging crop growth indices. Key examples of this approach include the use of the normalized difference vegetation index (NDVI) for crop monitoring and yield predictions in studies [6–9], as well as the application of the enhanced vegetation index (EVI) in research [10–12]. The integration of vegetative crop growth metrics (VGMs) derived from satellite data, as outlined by [5], into empirical crop yield models can significantly enhance the accuracy of yield estimates. This, in turn, supports more informed decision-making in crop production management.

In addition to the vegetation indices, weather dynamic changes significantly influence crop yield. Previous research has revealed the uncertainty of predicting crop yield with respect to crop growth stages, weather dynamic impacts, and model structures [13,14]. It has also been reported that climatic factors, such as temperatures and/or precipitation, could have impacts on the prediction of crop yields [15]. Spatial and temporal changes in temperature and precipitation have been analyzed in crop yield prediction [5,16,17]. However, these empirical models are specific to the crop variety, crop growth stages, and geographic locations [4,18]. To improve the accuracy of crop yield predictions by reducing uncertainty, it is necessary to combine these climatic factors in the prediction model for better yield estimation. For example, climatic factors are combined with a crop growth index (i.e., NDVI) using the Google Earth Engine to improve crop yield prediction [18].

In addition, several research efforts have been made to use different models to improve the yield prediction performance. For example, Beeri and Peled [19] proposed combining the real-time remotely sensed spatial distribution of vegetation with ground truth data for crop monitoring. Huang et al. [20] developed a new data assimilation cost function during key phenological stages to reduce errors in winter wheat yield estimation. It has also shown that maize yield prediction can be improved by combining soil moisture measurements and the leaf area index into a crop simulation model [13]. Machine learning, an important branch of artificial intelligence for learning from data, has also been widely used to predict yields based on different features and assumptions of the model [21]. For example, a deep Gaussian process has also been proposed to improve crop yield prediction [22]. Support vector machine classifiers have been trained to predict rice crop yields in India [23], and random forest regression models have been used to predict yield with climatic factors, using simulated biomass from the APSIM (Agricultural Production Systems Simulator) [24]. Spiking neural networks have been used for crop yield prediction based on spatiotemporal analysis of normalized difference vegetation index image time series [25]. More recently, motivated by the success of deep learning in computer vision, deep convolutional neural networks, which are highly nonlinear and scalable, have been used for improved crop yield prediction. For example, Wang et al. [26] leveraged deep transfer learning techniques to achieve promising results in predicting soybean crop yields in Argentina. Khaki et al. [27] developed a combination of convolutional neural networks (CNNs) and recurrent neural networks (RNNs), called CNN–RNN, for corn and soybean yield prediction based on environmental data and management practices. Srivastava et al. [28] proposed CNNs to predict the yield of winter wheat using crop phenology data and environmental data.

This study aims to analyze the yield of soybeans (one of the major crops in the Greater Mississippi River Basin—the largest agricultural region in the United States) from 2008 to 2021. It has been reported that soybean production has been increasing over the past decades. The use of remote sensing technology to model soybean yields could be a significant benefit to soybean management and planning, due to the low cost and availability of near real-time measurements. This study uses the NDVI, a data product of MODIS satellite images, with a 16-day interval at 250 m spatial resolution at a county level, to model and analyze the relationship between NDVI measurements at different days over the growing season and crop yields across different weather zones. Meanwhile, motivated by previous studies revealing the relationships between climatic factors and yields, this study also aims to answer the research question of how yields are correlated with different weather conditions in different weather regions. In particular, this study considers three types of weather variables, namely minimum temperature, maximum temperature, and precipitation, and analyzes how the crop yield can be impacted by these three climatic factors. At the end, we analyze correlations between climatic factors and yields for different weather regions and build predictive models using NDVI measurements and climatic factors to find out the best ones for soybean yield predictions. Our results reveal interesting correlation patterns for different weather regions, and that nonlinear models can achieve promising prediction performance.

2. Study Area

The study area for this research is the Greater Mississippi River Basin, where the majority of soybeans in the United States are planted. The Greater Mississippi River Basin, one of the largest river systems in the world, spans a significant portion of the continental United States. It covers approximately 1.2 million square miles, draining water from 31 states and 2 Canadian provinces. The basin extends from the Rocky Mountains in the west to the Appalachian Mountains in the east, and from the headwaters of the Mississippi River at Lake Itasca in Minnesota down to its mouth in the Gulf of Mexico. This vast watershed is divided into six major sub-basins: the Upper Mississippi, Lower Mississippi, Missouri, Ohio, Arkansas–White–Red, and Tennessee River basins.

The basin crosses four major distinct weather regions [29]: Very Cold, Cold, Mixed Humid, and Hot Humid. The Very Cold region, located primarily in the northern parts of the basin, experiences long, harsh winters and shorter growing seasons, impacting soybean growth. The Cold region, found just south of the Very Cold zone, has a somewhat milder weather dynamics but still features significant seasonal variation, with warm summers and cold winters. The Mixed Humid region, stretching across the central part of the basin, is characterized by moderate temperatures and balanced rainfall throughout the year, providing a favorable environment for soybean cultivation. Finally, the Hot Humid region, located in the southern part of the basin, experiences warm temperatures year-round with high humidity and ample rainfall, which can both benefit and challenge soybean production due to potential for excessive moisture and heat stress. Figure 1 shows the study area of the Greater Mississippi River Basin with its different weather regions, highlighting the mean annual flows from runoff across these diverse weather regions.

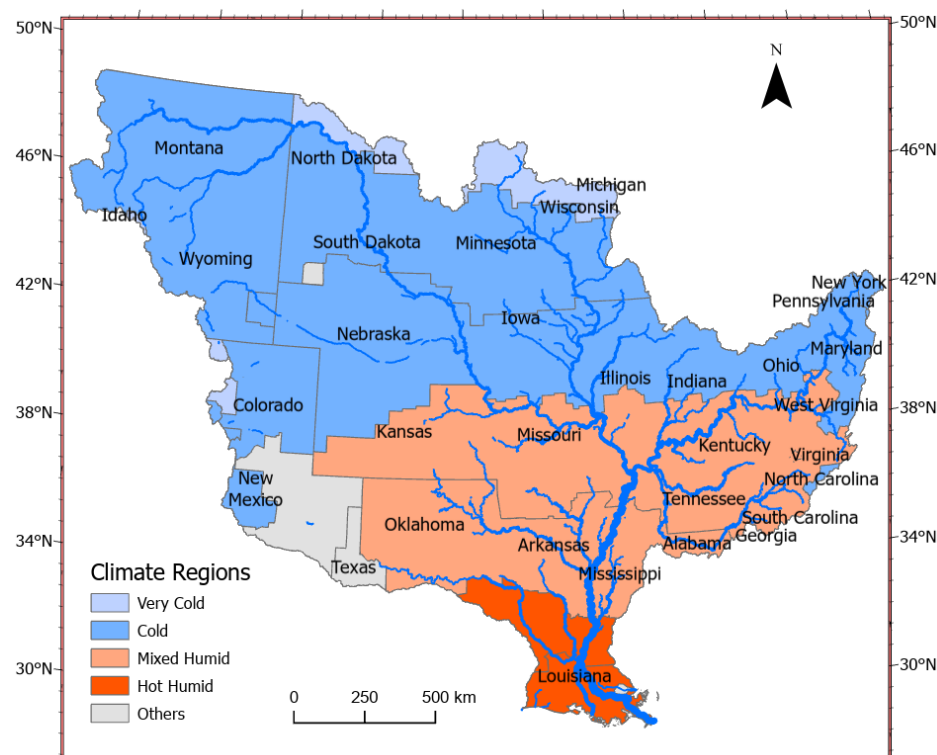


Figure 1. The study area of the Greater Mississippi Rivier Basin for soybean yield modeling.

3. Materials and Methods

3.1. Data Collection

This study used five publicly available data, including (i) the soybean cropland data layers that were collected from the USDA NASS (National Agriculture and Statistics Service, https://www.nass.usda.gov/Research_and_Science/Cropland/SARS1a.php: accessed on 15 March 2024) at 30 m spatial resolution, (ii) the soybean yield data that were collected and managed from the USDA NASS site (https://www.nass.usda.gov/Data_and_Statistics/: accessed on 15 March 2024) at the county level for each state at 30 m spatial resolution, (iii) the MODIS Vegetation Index Product–NDVI, which were continuous data products from MODIS at a 16-day interval and 250 m spatial resolution (<https://modis.gsfc.nasa.gov/data/dataproduct/mod13.php>: accessed on 20 March 2024), (iv) the environmental data, which were collected and managed from WorldClim (<https://www.worldclim.org/data/worldclim21.html>: accessed on 25 March 2024), and (v) the environmental zone layers, which were defined by the U.S. Department of Energy Building America Program at a county level, and the Greater Mississippi River Basin was further split into four major regions according to the weather zones: Very Cold, Cold, Mixed Humid, and Hot Humid. The soybean cropland data layers provide the boundaries of crop planting and harvesting areas at a high spatial resolution of 30 m × 30 m every year. For each soybean growing area, this study considered the NDVI crop growth metrics at different days to forecast the soybean crop yield for different weather zones. The used WorldClim database consists of monthly weather data, including minimum temperature, maximum temperature, and average precipitation, for global land areas at a spatial resolution of 30 s (~1 km²), which are produced from a large number of weather stations, interpolated using thin-plate splines with covariates including elevation, distance to the coast, and three satellite-derived covariates: maximum and minimum land surface temperature as well as cloud cover, and obtained with the MODIS satellite platform [30].

3.2. Geographic Data Processing

The overall schematic diagram for geographic data processing and yield modeling is shown in Figure 2. Geographic data processing consists of five major steps. Step 1 aims to extract the NDVI layer from the MODIS Vegetation Indices 16-day L3 Global 250 m data and multiply with the scale factor, which transforms the NDVI DN values into its standard range from 0 to 1. Notice that the soybean planting area data have a higher spatial resolution (30 m) than the remotely sensed NDVI images (250 m). It is necessary to first downscale the soybean planting area data, as shown in Step 2, to ensure that the whole monitored area of each extracted NDVI pixel (250 m × 250 m) is located within soybean fields. Then, we mask the NDVI layer with the soybean planting area for soybean crops to extract the corresponding vegetative growth indices. After that, Step 3 aims to summarize the NDVI values for each county in the study area, including mean, standard deviation, minimum, and maximum of NDVI. Similarly, Step 4 summarizes the weather data for each county, including the mean precipitation, the minimum temperature, and the maximum temperature. The soybean yield data are filtered to the counties with NDVI and weather data statistics. These steps were conducted for each of four weather zones and the county-level masked data were processed for crop yield modeling and the relationships between the crop yield and NDVI-based vegetative growth indices or weather variables.

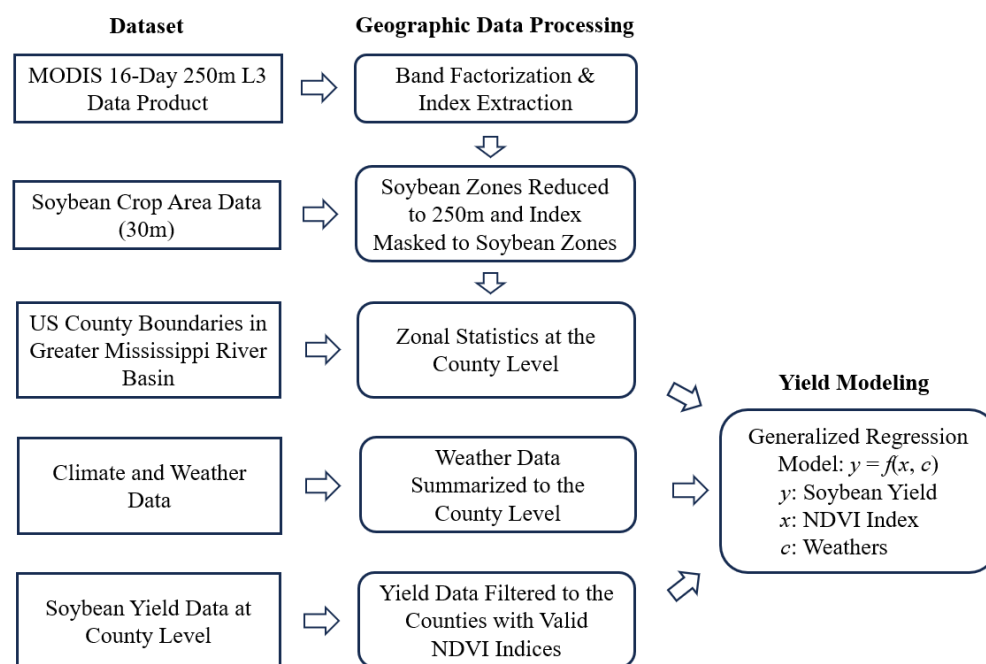


Figure 2. The schematic diagram for geographic data processing and yield modeling.

3.3. Correlation Analysis

Correlation analysis is applied in this study to understand the relationship between the soybean yield and the 16-day interval NDVI measurements, and the soybean yield and the monthly climatic factors. Specifically, for any two variables x and y with n measurements, we compute their Pearson's correlation coefficient [31] using the following formula:

$$\rho = \frac{cov(x, y)}{\sigma_x \sigma_y} = \frac{n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{\sqrt{n \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2} \sqrt{n \sum_{i=1}^n y_i^2 - (\sum_{i=1}^n y_i)^2}}$$

where $cov(x, y)$ denotes the covariance between x and y , and σ_x and σ_y denote the standard deviation of x and y , respectively. The value of Pearson's correlation coefficient ranges from -1 to $+1$. When the value comes down to zero, then the two variables are considered not

related. A value of +1 indicates that the two variables x and y are perfectly and positively correlated, and -1 shows a perfect negative correlation. In a weather region, this study evaluates the correlation for the following two types of variables: (i) Mean NDVIs (x) at a county level on different days and soybean yield (y). NDVI measurements are available every 16 days from Day 1, Day 17, Day 33, up to Day 353, and their correlation analysis can tell on which day the NDVIs have the highest correlation with the soybean crop yield under different weather regions. (ii) Monthly climatic factors (x) and soybean yield (y) at both the county level and the weather region level. Three climatic factors are considered individually, including mean precipitation, minimum temperature, and maximum temperature. Their correlation analysis can tell how the crop yield will be impacted by these three weather factors under different regions.

3.4. Regression Models

Linear regression models are first trained to predict crop yields based on NDVI-based vegetation metrics and climatic factors. In the linear model, only vegetation metrics are first considered as explanatory variables, and then climatic factors are added to illustrate the contribution of climatic factors for yield prediction. We summarize the linear regression models as below:

$$y = b + \sum_{i=1}^k a_i x_i^{NDVI}$$

$$y = b + \sum_{i=1}^k a_i x_i^{NDVI} + \sum_{i=1}^l c_i x_i^c$$

where y denotes crop yield; b is the intercept for the linear equation; x_i^{NDVI} denotes the i -th NDVI vegetation metric, and x_j^c denotes the j -th climatic factors. To avoid the multicollinearity for multivariate regression models, we adopt the stepwise regression method, which helps automatically find the best model to predict crop yield by adding or removing explanatory variables in a step-by-step process.

In addition to the intercept and linear term of each predictor, the nonlinear regression model further considers squared terms for each predictor and all products of pairs of distinct predictors. Mathematically, for m predictors, the nonlinear regression model is summarized as below:

$$y = b + \sum_{i=1}^m a_i x_i^{NDVI} + \sum_{i=1}^m c_i (x_i^{NDVI})^2 + \sum_{i \neq j} d_{ij} x_i^{NDVI} x_j^{NDVI}$$

$$y = b + \sum_{i=1}^m a_i x_i^{NDVI} + \sum_{i=1}^m c_i (x_i^{NDVI})^2 + \sum_{i \neq j} d_{ij} x_i^{NDVI} x_j^{NDVI} + \sum_{i=1}^m a_i x_i^c + \sum_{i=1}^m c_i (x_i^c)^2 + \sum_{i \neq j} d_{ij} x_i^c x_j^c$$

where b is the intercept, and a_i , c_i , and d_{ij} are coefficients for linear terms, squared terms, and their products of pairs, respectively. Although nonlinear relationships between the yield and all predictors are considered, the above equations are linear in their coefficients, and stepwise regression methods can be still applied to find the best model for crop yield prediction.

The best linear and nonlinear models are further compared with the support vector machine (SVM) model and random forest (RF) model. Both SVM and RF models are supervised machine learning algorithms that are widely used for classification and regression analysis. The SVM algorithm, which aims to find hyperplanes for the optimal separation of classes [32], was originally proposed for classification, and it was extended as a support vector regression (SVR) for prediction modeling [33]. The SVR used for crop yield modeling

first applies a gaussian kernel to transform the input space to a higher dimensional space. The hyperplane can be found by minimizing the following cost function:

$$|y_{combined} - \langle w, x \rangle - b| \leq \epsilon$$

where x is the training sample, w is normal vector to the hyperplane, and b is the intercept. The sum of inner product $\langle w, x \rangle$ and intercept is the predicted value of the sample within the range of ϵ .

The RF algorithm is a type of ensemble learning used for both classification and regression tasks. In the context of regression, an RF is built by training multiple decision trees on random subsets of the training data. Each tree in the forest is trained independently, and at each node of the tree, a random subset of features is considered for splitting. This randomness helps reduce overfitting, as it ensures that the trees in the forest are less correlated with each other. For a regression task, the prediction from an RF model is obtained by averaging the predictions from all the individual trees. This aggregation of multiple decision trees helps improve the model's robustness and generalization ability, especially when dealing with high-dimensional data or complex relationships between variables. Mathematically, let $T_1(x)$, $T_2(x)$, ..., $T_n(x)$ represent the predictions from n individual trees in the forest. For our crop yield predictions, $n = 100$ trees are trained and aggregated for the final prediction. The final prediction, denoted by $\hat{y}$, can be found using the following equation:

$$\hat{y} = \frac{1}{n} \sum_{i=1}^n T_i(x)$$

Once a model is trained, the model performance will be evaluated to find the best model for crop yield prediction. This study will use the following widely used performance metrics: the adjusted R-squared, root mean square error (RMSE), and normalized root mean square error (NRMSE). The adjusted R^2 penalizes models with more independent variables compared to the regular R^2 , and hence it is a more reliable metric for comparing models with different numbers of predictors. The mathematical equations for the above performance metrics are summarized below:

$$\text{Adjusted } R^2 = 1 - \frac{(N-1)SSE}{(N-P)SST}$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$\text{NRMSE} = \frac{\text{RMSE}}{1/n \sum_{i=1}^n \hat{y}_i}$$

where $SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ is the sum of squares of residuals called the residual (error) sum of squares, $SST = \sum_{i=1}^n (y_i - \bar{y})^2$ is the total sum of squares, and $\bar{y}$ is the mean value of y .

4. Results and Discussions

4.1. Variable Statistics

We first summarize the temporal statistics of our variables of interest, including soybean yields, vegetative metrics, and climatic factors, over time.

Soybean Yield: The average soybean yield (bushel/acre) of each weather region over the years is shown in Figure 3, where bushel is a measure of capacity equal to 64 US pints (equivalent to 35.2 L). It can be shown that the Cold region generally has a higher soybean yield than the other weather regions, except for the years of 2012 and 2014, when the Hot

Humid region had a significant yield increase. The Hot Humid and Mixed Humid regions have similar yield performances, and the Very Cold region generally has the lowest soybean yield compared to the other weather regions.

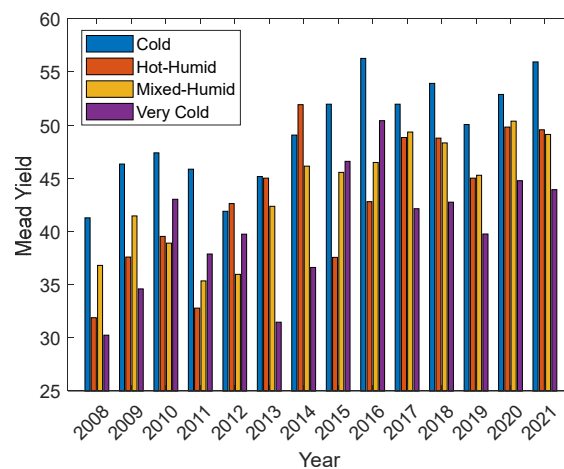


Figure 3. Average soybean yield of four environmental regions over the years.

Vegetative Growth Metric and Weather Factors: The NDVI measurements provide a quantitative assessment of vegetation health and density by utilizing red and near-infrared spectral bands, which vary throughout the crop season. Climatic factors also play important roles in the growth of soybeans, with a beginning planting date in early April and a last harvesting date in late October, as reported by the USDA National Agricultural Statistics Service [34]. Hence, it is important to understand the temporal changes in the NDVI-based vegetative growth metrics and climatic factors over the soybean growth period. For each weather region, the maximum NDVI observed over its soybean growth season, called VGM_{max} , is used as the effective vegetative metric for soybean modeling [5]. The VGM_{max} and corresponding climatic factors, including mean precipitation, minimum temperature, and max temperature, for different weather regions are shown in Figure 4a–d, respectively. Figure 4a illustrates that the Cold region generally exhibits the highest VGM_{max} , while the Hot Humid region shows the lowest VGM_{max} . Additionally, significant declines in VGM_{max} were observed in 2012 for both the Cold and Mixed Humid regions, and in 2015 for the Hot Humid region. Figure 4b reveals a consistent temporal pattern of mean precipitation over years across all four weather regions. Figure 4c shows distinct differences in minimum temperatures among the regions, with the Hot Humid region having the highest minimum temperatures and the Very Cold region having the lowest minimum temperatures. It is also notable that there was a significant decrease in minimum temperatures in 2012 for the Mixed Humid, Cold, and Very Cold regions. Similarly, Figure 4d also shows distinct differences in maximum temperatures across the four weather regions.

4.2. Correlation Analysis

The correlation analysis between the yield and four explanatory variables, including VGM_{max} , mean precipitation, maximum temperature, and minimum temperature, at the county level for different weather regions are shown in Figure 5. It clearly shows positive correlations between the vegetative growth metric and the yield for all four weather regions. Regarding climatic factors, we find that the mean precipitation is generally positively correlated with the yield in the four different weather regions; however, negative correlations with the yield are observed for the maximum temperature in the Hot Humid and Mixed Humid regions and for the minimum temperature in the Cold and Very Cold regions.

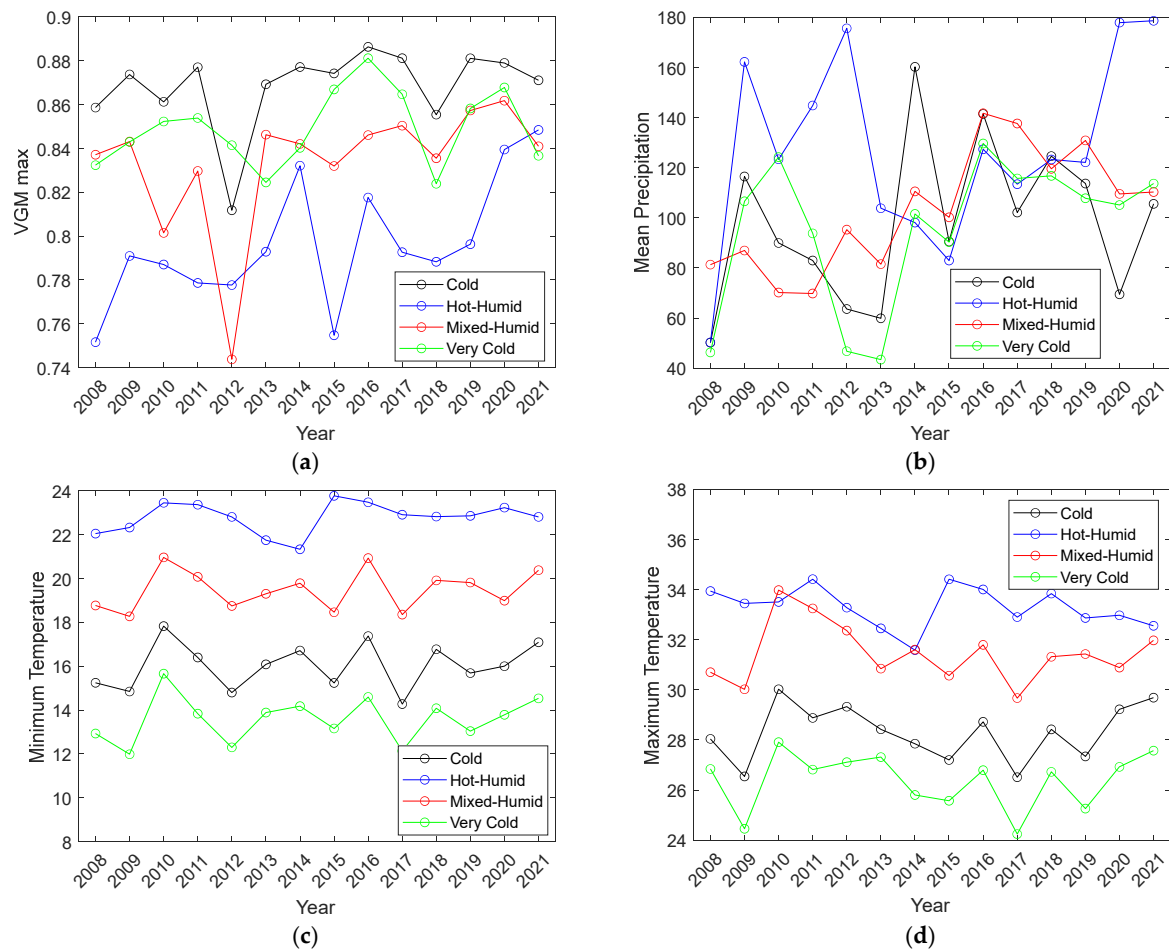


Figure 4. The temporal pattern of VGM_{max} (a) and three corresponding climatic factors: mean precipitation (b), minimum temperature (c), and maximum temperature (d).

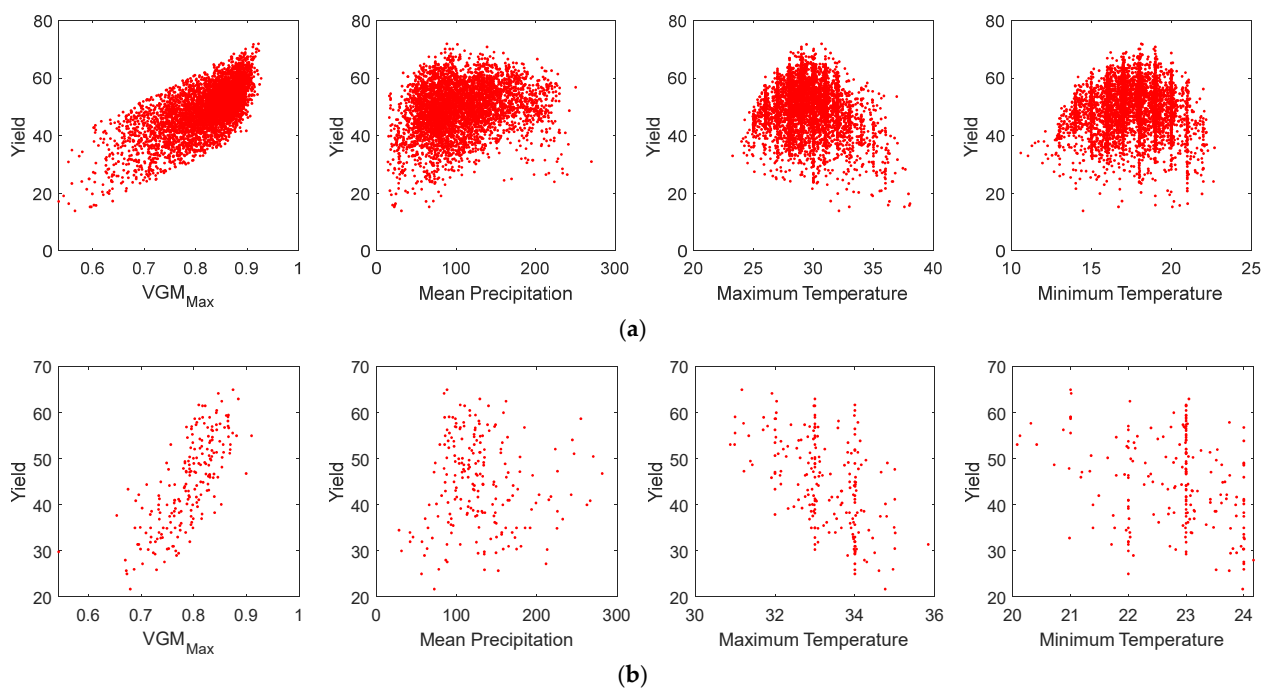


Figure 5. Cont.

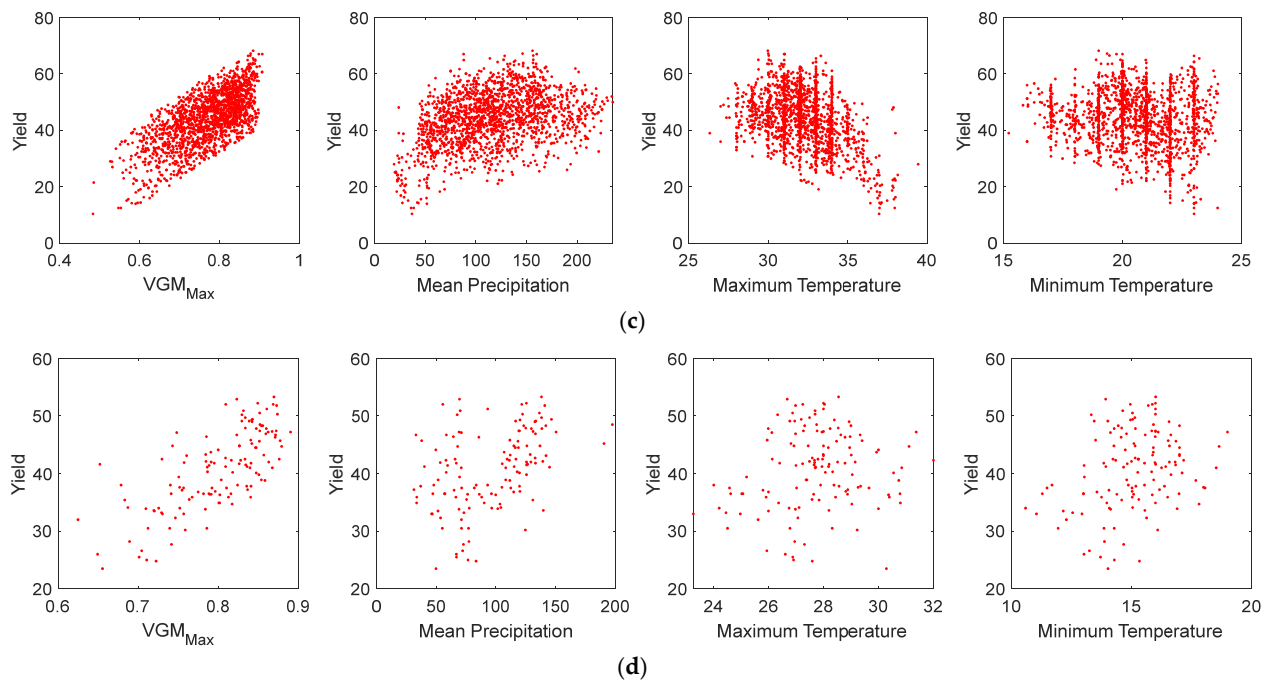


Figure 5. The relationship between the yield and four explanatory variables: VGM_{max} , mean precipitation, maximum temperature, and minimum temperature at the county level in (a) Cold, (b) Hot Humid, (c) Mixed Humid, and (d) Very Cold regions.

We also noted the temporal relationship between the yield and all individual explanatory variables at the weather region level. The temporal relationship between the vegetative growth metric VGM_{max} and the yield is illustrated in Figure 6. Both VGM_{max} and yield have been normalized to a range of [0–1] to facilitate a clearer understanding of their temporal relationship. Generally, a positive correlation is observed across all weather regions. To further analyze this relationship, we applied local correlation tracking [35] to identify the periods of highest correlation without normalization at two temporal scales: 5 years and 10 years. Specifically, for the 10-year scale, we computed Pearson’s correlation coefficient for every 10 consecutive years and identified the period with the highest coefficient. The same procedure was applied for the 5-year scale. The highest correlation coefficients, along with their corresponding years, are summarized in Table 1 for four different weather regions, at both temporal scales. The results indicate significant correlations between the vegetative growth metric VGM_{max} and the yield across all weather regions, suggesting that VGM_{max} could be an effective factor to predict the yield. Notably, in the Very Cold region, we observed a Pearson’s correlation coefficient of 0.935 for the period from 2008 to 2017 at the 10-year scale, and a coefficient of 0.988 for the period from 2013 to 2017 at the 5-year scale.

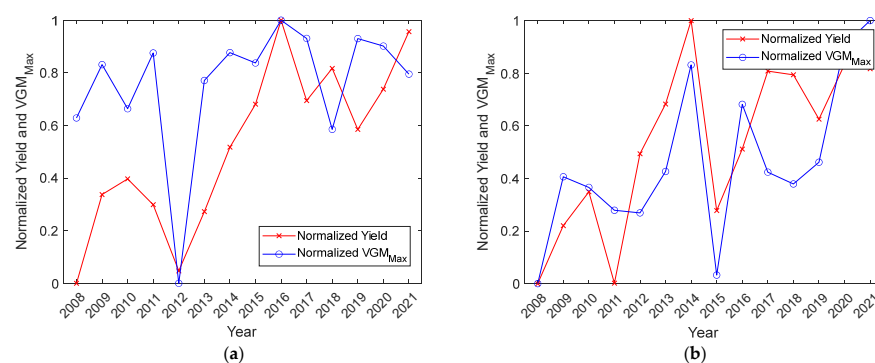


Figure 6. Cont.

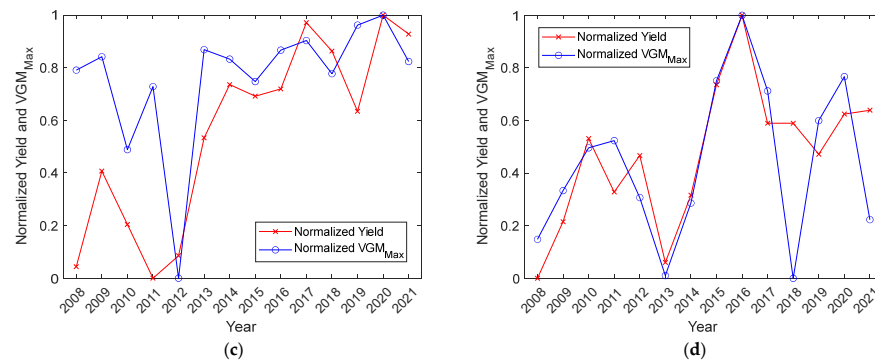


Figure 6. The temporal relationship between VGM_{max} and yield, both normalized, over time in Cold (a), Hot Humid (b), Mixed Humid (c), and Very Cold (d) regions.

Table 1. Local temporal correlation analysis between the yield and VGM_{max} at two scales: 10 years and 5 years. The highest correlation coefficients and corresponding years are given for four different weather regions.

Weather Regions	Maximum Local Correlation (10-Year Scale)	Maximum Local Correlation (5-Year Scale)
Cold	0.686 (2008–2017)	0.909 (2013–2017)
Hot Humid	0.735 (2008–2017)	0.830 (2010–2014)
Mixed Humid	0.823 (2012–2021)	0.935 (2012–2016)
Very Cold	0.935 (2008–2017)	0.988 (2013–2017)

Additionally, we explored the correlations between yield and various weather factors across different weather regions over time. We summarized the local temporal correlation analysis results at both 5-year and 10-year scales in Table 2, which lists the highest correlation coefficients and their corresponding years for four distinct weather regions. The results in Table 2 reveal interesting relationships between the yield and the climatic factors across various weather regions. Overall, yield shows a positive temporal correlation with precipitation across all regions. In the Cold and Very Cold regions, yield exhibits a positive temporal correlation with the minimum temperatures, suggesting that yield tends to increase with higher temperatures in these areas. Conversely, in the Hot Humid and Mixed Humid regions, yield has a negative temporal correlation with the maximum temperature, indicating that yield tends to increase with lower temperatures in these areas. These relationships provide valuable insights into the impact of weather change on soybean yield, which could be crucial for future agricultural planning and adaptation strategies.

Table 2. Ten years and five years. The highest correlation coefficients and corresponding years are given for four different weather regions.

Weather Regions	Yield and Climatic Factors	Maximum Local Correlation (10-Year Scale)	Maximum Local Correlation (5-Year Scale)
Cold	Yield vs. Precipitation	0.699 (2009–2018)	0.970 (2009–2013)
	Yield vs. Max Temp	0.175 (2008–2017)	0.912 (2013–2017)
	Yield vs. Min Temp	0.600 (2012–2021)	0.944 (2014–2018)

Table 2. Cont.

Weather Regions	Yield and Climatic Factors	Maximum Local Correlation (10-Year Scale)	Maximum Local Correlation (5-Year Scale)
Hot Humid	Yield vs. Precipitation	0.444 (2010–2019)	0.590 (2010–2014)
	Yield vs. Max Temp	−0.845 (2011–2020)	−0.964 (2011–2015)
	Yield vs. Min Temp	−0.606 (2012–2021)	−0.935 (2016–2020)
Mixed Humid	Yield vs. Precipitation	0.663 (2012–2021)	0.771 (2012–2016)
	Yield vs. Max Temp	−0.408 (2012–2021)	−0.779 (2016–2020)
	Yield vs. Min Temp	0.277 (2008–2017)	−0.781 (2016–2020)
Very Cold	Yield vs. Precipitation	0.360 (2008–2017)	0.428 (2008–2012)
	Yield vs. Max Temp	0.717 (2008–2017)	0.939 (2013–2017)
	Yield vs. Min Temp	0.868 (2008–2017)	0.966 (2012–2016)

4.3. Yield Modeling and Prediction

For each weather region, we created two models: y_{NDVI} and y_{combined} , where y_{NDVI} is a linear regression model using only the NDVI-based vegetative growth metric VGM_{max} as the explanatory variable, and y_{combined} is a linear regression model using both VGM_{max} and three climatic factors as the explanatory variables. Comparing the performance of these two models can illustrate the contribution of climatic factors for yield prediction. Table 3 summarizes the best linear models for different weather regions using the stepwise regression method. The stepwise linear regression model uses the Bayesian information criterion to add or remove linear terms. Based on the model coefficients, we found that, for all weather regions, the vegetative growth metric has a positive impact on yield prediction. Regarding the climatic factors, the coefficients of the regression model using all years' data indicate the positive impacts of precipitation and maximum temperature on yield prediction for the Cold and Very Cold regions, while negative coefficients of maximum temperature are found for the Hot Humid and Mixed Humid regions. These results indicate that increased temperature and precipitation in the Cold and Very Cold regions could help to increase the yield, whereas this could decrease the yield for the Hot and Mixed Humid regions. We also applied the stepwise regression technique to the quadratic nonlinear regression model to find the best terms, as summarized in Table 4. Compared to the best stepwise linear models, all weather regions, except for the Very Cold region, include the squared terms and the product terms of some variables. Table 5 summarizes the performance of the best-predicted linear models, compared with the quadratic nonlinear regression, SVM regression, and RF regression methods. These nonlinear models use complex nonlinear models, and they have the potential to better fit the data with improved prediction performance. The model results show the same reliability for the best linear model y_{combined} (adjusted R-squared of 0.57, 0.61, 0.57, and 0.59) and the RF regression model (adjusted R-squared of 0.84, 0.84, 0.85, and 0.82) for Cold, Hot Humid, Mixed Humid, and Very Cold regions. The prediction performance of the nonlinear model is generally better than the linear model in terms of the adjusted R-squared, RMSE, and NRMSE values. The SVM regression model achieves the best performance for the Hot Humid region (adjusted R-squared of 0.93) and Very Cold region (adjusted R-squared of 0.91), and the RF regression model performs the best for the

Cold region (adjusted R-squared of 0.84) and Mixed Humid region (adjusted R-squared of 0.85).

Table 3. Linear yield prediction model for each weather region for soybean crops.

Weather Region	Best Linear Models
Cold	$y_{\text{combined}} = 161.76 \times \text{VGM}_{\text{max}} + 0.01 \times \text{Precipitation} + 0.14 \times \text{MaxT} - 95.94$
Hot Humid	$y_{\text{combined}} = 113.30 \times \text{VGM}_{\text{max}} - 2.67 \times \text{MaxT} + 43.80$
Mixed Humid	$y_{\text{combined}} = 65.72 \times \text{VGM}_{\text{max}} - 2.84 \times \text{MaxT} + 2.22 \times \text{MinT} + 39.13$
Very Cold	$y_{\text{combined}} = 84.42 \times \text{VGM}_{\text{max}} + 0.04 \times \text{Precipitation} + 0.06 \times \text{MaxT} - 39.10$

Note: All models have a p -value of less than 0.00001.

Table 4. Quadratic nonlinear yield prediction model for each weather region for soybean crops.

Weather Region	Best Quadratic Models
Cold	$y_{\text{combined}} = -858.23 \times \text{VGM}_{\text{max}} + 0.19 \times \text{Precipitation} - 21.50 \times \text{MaxT} + 51.63.50 \times \text{MinT}$ $+ 24.87 \times \text{VGM}_{\text{max}} \times \text{MaxT} - 36.51 \times \text{VGM}_{\text{max}} \times \text{MinT} - 1.45 \times \text{MaxT} \times \text{MinT}$ $+ 525.82 \times \text{VGM}_{\text{max}}^2 + 0.43 \times \text{MaxT}^2 + 0.65 \times \text{MinT}^2 + 211.48$
Hot Humid	$y_{\text{combined}} = -1031.10 \times \text{VGM}_{\text{max}} - 19.03 \times \text{MaxT} + 20.84 \times \text{VGM}_{\text{max}} \times \text{MaxT} + 292.69 \times \text{MaxT}^2 + 761.57$
Mixed Humid	$y_{\text{combined}} = -385.53 \times \text{VGM}_{\text{max}} - 0.36 \times \text{Precipitation} - 30.85 \times \text{MaxT} + 47.52 \times \text{MinT}$ $+ 0.52 \times \text{VGM}_{\text{max}} \times \text{Precipitation} + 21.66 \times \text{VGM}_{\text{max}} \times \text{MaxT} - 23.96 \times \text{VGM}_{\text{max}} \times \text{MinT}$ $- 2.81 \times \text{MaxT} \times \text{MinT} + 136.90 \times \text{VGM}_{\text{max}}^2 + 1.05 \times \text{Precipitation}^2 + 1.59 \times \text{MinT}^2 + 207.19$
Very Cold	$y_{\text{combined}} = 84.42 \times \text{VGM}_{\text{max}} + 0.04 \times \text{Precipitation} + 0.06 \times \text{MaxT} - 39.10$

Note: All models have a p -value of less than 0.00001.

Table 5. Comparison of model prediction performances between linear and nonlinear models for each weather region for soybean crops.

Models	Performance	Cold	Hot Humid	Mixed Humid	Very Cold
Linear Regression	Adj R ²	0.57	0.61	0.57	0.59
	RMSE	5.68	6.22	6.20	4.43
	NRMSE	0.11	0.12	0.14	0.10
Nonlinear Quadratic Regression	Adj R ²	0.58	0.68	0.60	0.64
	RMSE	5.45	5.52	5.91	4.13
	NRMSE	0.11	0.12	0.13	0.10
SVM Regression	Adj R ²	0.68	0.93	0.82	0.91
	RMSE	4.78	2.50	3.99	2.00
	NRMSE	0.10	0.06	0.09	0.05
RF Regression	Adj R ²	0.84	0.84	0.85	0.82
	RMSE	3.43	3.89	3.67	2.96
	NRMSE	0.07	0.09	0.08	0.07

While the nonlinear regression model can typically fit the data better than the linear regression model, overfitting could happen with nonlinear regression models, which prevent their applicability to predict future crop yields. To find the best model for the actual field predictability, we conducted cross validation for both the linear and nonlinear models. Specifically, we randomly chose 20% of the data as the test data and the remaining 80% of the data as the training data. For the training dataset, we applied 4-fold cross validation for hyperparameter tuning. By doing so, the test data are not seen when training the model, which can offer a more robust estimate of how well the model generalizes to unseen data. Table 6 summarizes the cross-validated model prediction performances between linear and nonlinear models for each weather region. The results show that the cross-validated model

has a reduced prediction performance compared with the one without cross validation shown in Table 5, indicating that the prediction model without cross validation may have an overfitting issue to some extent. This study found that cross-validated SVM regression performs the best in the Mixed Humid region in terms of the adjusted R-squared (0.67) and in the Very Cold region (0.69), and that the cross-validated RF regression model achieves the best adjusted R-squared in the Hot Humid region (0.61) and Cold region (0.62), compared to all other models. For the three nonlinear models, they achieve similar cross validation performances for crop yield prediction. In general, the nonlinear cross-validated model can achieve better prediction performance than the linear cross-validated model over all four weather regions.

Table 6. Comparison of cross-validated model prediction performances between linear and nonlinear models for each weather region for soybean crops.

Models	Performance	Cold	Hot Humid	Mixed Humid	Very Cold
Linear Regression	Adj R ²	0.54	0.55	0.57	0.62
	RMSE	5.91	6.62	6.12	4.25
	NRMSE	0.12	0.14	0.14	0.12
Nonlinear Quadratic Regression	Adj R ²	0.60	0.60	0.61	0.68
	RMSE	5.52	5.85	5.72	3.81
	NRMSE	0.11	0.12	0.13	0.09
SVM Regression	Adj R ²	0.61	0.60	0.67	0.69
	RMSE	5.42	6.10	5.20	3.75
	NRMSE	0.11	0.13	0.12	0.09
RF Regression	Adj R ²	0.62	0.61	0.66	0.58
	RMSE	5.35	5.76	5.28	4.52
	NRMSE	0.11	0.12	0.12	0.10

5. Discussions

This study modeled soybean crop yield for different weather regions in the Greater Mississippi River Basin and analyzed the impact of climatic factors, including precipitation, minimum temperature, and maximum temperature, on the yield. Our findings confirmed that the NDVI-based vegetative growth metric is a key predictor of soybean yield, consistent with previous reports by [5,18]. In addition, this study identified the significant impacts of climatic factors on soybean yield modeling. Compared to the work [18], our study considered weather zones as defined by the U.S. Department of Energy, based on widely accepted classifications of world weathers that have been applied in a variety of different disciplines, and our correlation analysis revealed important and distinct temporal relationships between climatic factors and soybean yields for different weather regions. For example, while a consistent positive temporal correlation between precipitation and yield is found across all weather regions, for the Cold and Very Cold regions, yield exhibits a positive temporal correlation with the minimum temperature, suggesting that yield tends to increase with warmer weather in these areas, and for the Hot Humid and Mixed Humid regions, yield has a negative temporal correlation with the maximum temperature, indicating that yield tends to increase with cooler weather in these areas. These relationships provide valuable insights into the impact of weather dynamics on soybean yield, which could be crucial for future agricultural planning and adaptation strategies. Meanwhile, supported by these correlation analysis findings, this approach leads to more reliable cross-validated yield predictions.

Accurate crop yield prediction has garnered significant attention in recent years, with efforts focused on finding better models by comparing yield prediction accuracy, cost-effectiveness, reliability, and applicability. In line with previous crop yield modeling studies [6,12,15,20,36], this study developed crop yield prediction models that account for the variability of climatic factors across different regions and estimated the importance of these predictors. The accuracy of data-driven crop yield prediction models generally depends on factors such as crop growth phenology and weather data, as well as the resolution and accuracy of ground data. Consistent with prior research [5,18], this study utilized NDVI measurements as vegetative growth metrics and demonstrated effective performance in soybean yield prediction. Additionally, our correlation analysis and yield modeling results highlight the significant relationships between climatic factors and yield.

While it has been well reported that the impact of weathers on crop yield varies based on geographic location [18,37–39], this study found a positive impact of precipitation on soybean yield across all weather regions. Furthermore, we observed a positive impact of temperature on yield in the Cold and Very Cold regions, and a negative impact of temperature on yield in the Hot Humid and Mixed Humid regions. These findings provide valuable insights for predicting crop yields amid the growing risks associated with weather dynamics.

Compared to the study [15], our linear soybean yield prediction model demonstrated an improved adjusted R-squared between 0.57 and 0.61. Moreover, our linear model achieved an RMSE ranging from 4.43 to 6.22, which is lower than those reported by [8,18,40] for crop yield forecasting. Additionally, while nonlinear models such as SVM regression and RF regression further enhanced prediction accuracy compared to the linear model, as evidenced by the improvements in adjusted R-squared, RMSE, and NRMSE values, our study further shows that nonlinear models usually suffer from the overfitting issue much more than the linear model, as seen with the prediction performance drop in the cross-validated nonlinear models, which are tested by unseen data that are not included in the training dataset. Our study shows that the non-cross-validated nonlinear models, e.g., the random forest approach, can achieve the best prediction performance (see Table 5) across all weather regions, due to its combination of multiple prediction models, while the cross-validated linear model shows a relatively smaller performance decrease compared to the non-cross-validated linear model (see Table 6).

6. Conclusions

This study developed crop yield prediction models using remotely sensed data and additional climatic factors, focusing on different weather regions within the Greater Mississippi River Basin, USA. The research aimed to address the critical question of how climatic factors influence soybean yield across diverse weather regions, a topic of increasing importance in the face of weather dynamics. By identifying significant regional variations in climatic impacts, the study provides valuable insights into how specific factors—such as temperature and precipitation—affect crop productivity. For example, a higher minimum temperature was found to enhance yield in the northern regions (Cold and Very Cold), while lower maximum temperatures were more favorable for yield in the central and southern regions (Hot Humid and Mixed Humid). Precipitation, positively correlated with yield across all regions, emerged as a key factor in understanding soybean productivity.

This research was conducted to bridge the gap in knowledge regarding region-specific weather–yield interactions and to develop predictive models that can guide data-driven agricultural decision-making. Both linear and nonlinear regression models demonstrated strong predictive reliability, with the lowest NRMSE values observed in the Very Cold (0.10) and Cold (0.11) regions, and slightly higher but comparable values (0.12) in the Hot Humid

and Mixed Humid regions, using the best cross-validated random forest (RF) regression model. These results underscore the potential of integrating remotely sensed and climatic data for cost-effective and scalable agricultural modeling.

By providing actionable insights into weather–crop relationships, this study contributes to advancing agricultural management practices and building resilience against weather variability. The outcomes of this research are particularly relevant for farmers, policymakers, and researchers, enabling the development of region-specific strategies to enhance soybean yield, optimize resource use, and support sustainable food production. Ultimately, this work strengthens the scientific foundation for adapting agriculture to a changing weather dynamics, benefiting both the agricultural community and society at large.

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